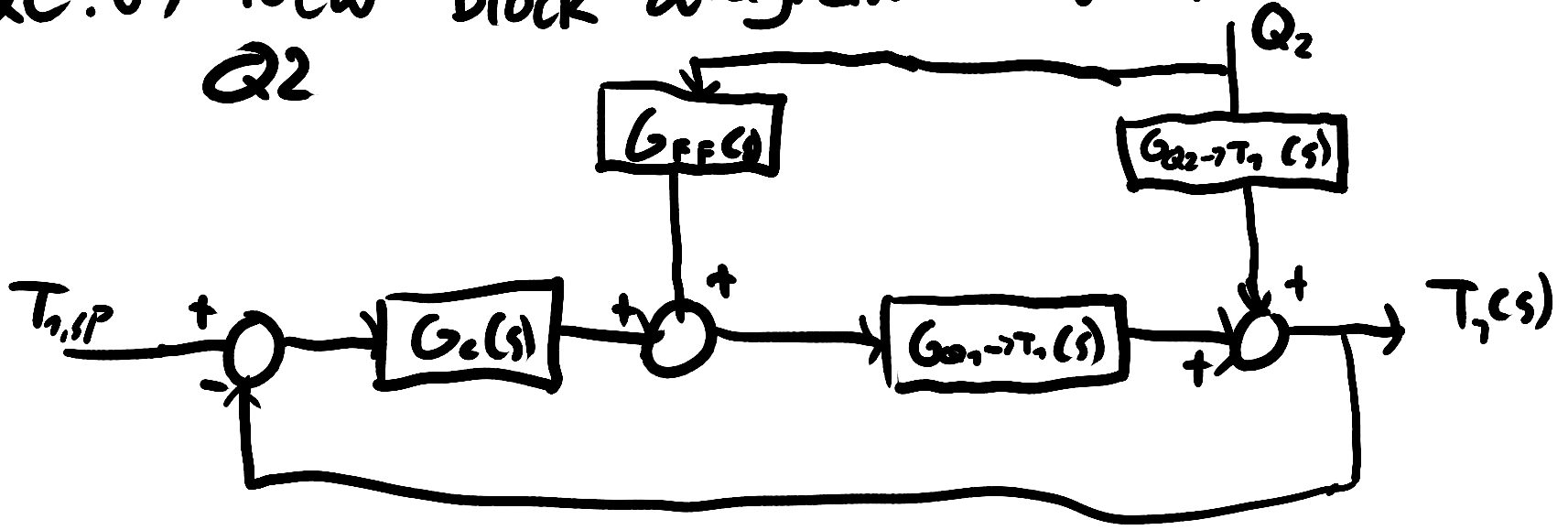
QC.6) New block diagram with feed forward for Q2



$$G_{FF}(s) = -\frac{G_{Q_2\to T_1}(s)}{G_{Q_1\to T_1}(s)}$$

$$= -\frac{K_D}{K_P}\cdot\frac{\frac{1}{\tau_{D}s+1}}{\frac{1}{\tau_P s+1}}\cdot\frac{e^{-\theta_D s}}{e^{-\theta_P s}}$$

$\tau_D = 14.2s$
$\tau_P = 15.9s$
} similar, so we assume the dynamic term to be 1.

$\theta_D = 24s$
$\theta_P = 6s$
} 18 second time difference. FF controller needs to wait 18s before applying control procedure against Q_2. It controls just by using $-\frac{K_D}{K_P}$.

$$-\frac{K_D}{K_P} = -0.538$$